

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416463
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Hartman; Seth D.

Helical electromagnetic counter recoil mechanism

Abstract

An artillery configured to propel a projectile, the artillery comprising a firing mechanism, a barrel, and a helical electromagnetic recoil mechanism. The firing mechanism fires the projectile in a first direction toward a target. The barrel has an open-ended bore that guides the projectile in the first direction. The helical electromagnetic counter recoil mechanism includes an electrical energy source, a stator, and a helical electromagnetic armature. The electrical energy source provides electrical energy when the firing mechanism fires the projectile. The stator and armature generate an electromagnetic force via the electrical energy. The helical electromagnetic armature moves in a second direction opposite the first direction via the electromagnetic force when the firing mechanism fires the projectile in the first direction to minimize recoil of the artillery.

Inventors: Hartman; Seth D. (Belton, MO)

Applicant: Honeywell Federal Manufacturing & Technologies, LLC (Kansas City, MO)

Family ID: 1000006861099

Assignee: Honeywell Federal Manufacturing & Technologies, LLC (Kansas City, MO)

Appl. No.: 17/667601

Filed: February 09, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 17361407 20210629 US 12196507 child-doc US 17667601

Publication Classification

Int. Cl.: F41A21/36 (20060101); H02K1/12 (20060101); H02K1/34 (20060101); H02K7/04 (20060101); H02K7/14 (20060101)

U.S. Cl.:

CPC **F41A21/36** (20130101); **H02K1/12** (20130101); **H02K1/34** (20130101); **H02K7/04** (20130101); **H02K7/14** (20130101);

Field of Classification Search

CPC: F41A (21/36); H02K (1/12); H02K (1/34); H02K (7/04); H02K (7/14)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3296928	12/1966	Lubbers	102/444	F41A 33/00
4369759	12/1982	Gerstenberger	124/68	F41A 3/78
4527457	12/1984	Fikse	318/135	F41B 6/006
5074189	12/1990	Kurtz	89/135	F41A 19/61
6694856	12/2003	Chen	89/43.01	F16F 13/007
8686576	12/2013	Smith	290/1R	F03G 7/08
10082360	12/2017	Hartman et al.	N/A	N/A
10823523	12/2019	Wynes	N/A	F41A 25/04
10976129	12/2020	Timpson	N/A	F41B 6/003
11150047	12/2020	Griffin	N/A	N/A
11209229	12/2020	Perez	N/A	N/A
11469025	12/2021	Hartman et al.	N/A	N/A
2012/0137870	12/2011	Lindsay	42/1.06	F41A 1/10
2015/0226507	12/2014	Palmer	89/1.703	F41A 1/08
2017/0343314	12/2016	Hartman	N/A	F41B 6/003
2021/0025669	12/2020	Griffin	N/A	F41C 23/06
2021/0102768	12/2020	Perez	N/A	F41A 3/84
2021/0183555	12/2020	Hartman	N/A	H02K 5/167
2022/0065564	12/2021	Kirstein	N/A	F41A 3/46

OTHER PUBLICATIONS

T.G. Engel, High-Efficiency, Medium-Caliber Helical Coil Electromagnetic Launcher, Nov. 2005, IEEE Transactions on Magnetics, vol. 41, No. 11, pp. 4299-4303 (Year: 2005). cited by examiner
Non-Final Office Action mailed Nov. 15, 2022 in related U.S. Appl. No. 17/361,407, 12 pages.
cited by applicant

Primary Examiner: Lopez; Michelle

Attorney, Agent or Firm: Hovey Williams LLP

Background/Summary

RELATED APPLICATION (1) The present patent application is a continuation-in-part (CIP) patent application claiming priority benefit, with regard to all common subject matter, to U.S. patent application Ser. No. 17/361,407, entitled “HELICAL ELECTROMAGNETIC COUNTER

RECOIL MECHANISM”, filed Jun. 29, 2021. The earlier-filed patent application is hereby incorporated by reference in its entirety into the present application.

BACKGROUND

(1) Firing large projectiles induces significant recoil to artilleries, which can affect accuracy and can damage artillery components and structures. Gas piston counter recoil mechanisms are limited by fluid dynamics and have a slight delay, which results in some unmitigated recoil. Gas piston counter recoil mechanisms also have a center of mass offset from the projectile's firing axis, which causes unmitigated pitching. Recoilless rifles expel exhaust to reduce recoil, but they cannot be used in tanks and other situations.

(2) Other devices that experience recoil suffer from similar problems. For example, jackhammers generate significant vibrations, making them difficult to control and causing significant user fatigue. Firearms are subject to recoil when fired, making them difficult to aim for subsequent shots and also causing significant user fatigue.

SUMMARY

(3) Embodiments of the invention solve the above-mentioned problems and other problems and provide a distinct advancement in the art of projectile propulsion recoil mitigation. More particularly, the invention provides a projectile propulsion device including a helical electromagnetic counter recoil mechanism that converts stored electrical energy to kinetic energy to counter recoil.

(4) An embodiment of the invention is an artillery broadly comprising a base, a firing mechanism, a barrel, and a helical electromagnetic counter recoil mechanism having an external configuration. The artillery may be a tank gun, an airplane gun, a battleship gun, or the like. Other embodiments may encompass projectile propulsion devices more generally such as an electromagnetic launcher, a gas gun, a machine gun, or the like.

(5) The base supports or houses the firing mechanism and barrel and may include a carriage, a gun mount, a firing platform, an elevator, and the like. The base may be maneuverable about a vertical axis (azimuth) and may be configured to pitch the barrel about a horizontal axis (altitude).

(6) The firing mechanism activates an explosive charge of a projectile. The firing mechanism may include a firing pin, a firing spring, a trigger, electrical or mechanical actuators, or the like.

(7) The barrel extends forward from the firing mechanism and forms an open-ended bore. The barrel is configured to guide the projectile out of the open end of the bore towards a target. The barrel may include rifling, muzzle brakes, propellant gas ports, and the like.

(8) The helical electromagnetic counter recoil mechanism includes an electrical energy source, a positive rail, a negative rail, a number of brushes, a stator, and a helical electromagnetic armature. The helical electromagnetic counter recoil mechanism at least partially encircles the barrel so as to be external relative to the barrel.

(9) The electrical energy source provides electrical energy to at least one of the stator and helical electromagnetic armature. The electrical energy source may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources.

(10) The positive rail electrically connects the stator to the electrical energy source via one of the brushes. Similarly, the negative rail electrically connects the helical electromagnetic armature to the electrical energy source via another one of the brushes.

(11) The stator includes helical coils and is stationary relative to the helical electromagnetic armature. The stator may be longitudinally aligned with the barrel. The stator may also encircle at least a portion of the barrel for more compact construction. The stator is electrically connected to the positive rail via one of the brushes and to the helical electromagnetic armature via another one of the brushes.

(12) The helical electromagnetic armature includes helical coils and is configured to longitudinally translate relative to the stator. The helical electromagnetic armature may encircle at least a portion

of the stator and may encircle at least a portion of the barrel. The helical electromagnetic armature is electrically connected to the stator via one of the brushes and to the negative rail via one of the brushes.

(13) In use, the artillery receives a reaction force realized in the form of recoil due to rapid forward acceleration of the projectile when the projectile is fired. The recoil is minimized or eliminated by the helical electromagnetic counter recoil mechanism. Specifically, the electrical energy source provides electrical energy in the form of electrical current to the stator and helical electromagnetic armature. Current passing through the coils of the stator and/or the coils of the helical electromagnetic armature creates an electromagnetic force that moves the helical electromagnetic armature in a second direction opposite the first direction at substantially the same time the projectile is propelled in the first direction. The artillery receives a reaction force from the helical electromagnetic armature that counters the reaction force or recoil imparted on the artillery by the projectile. The reaction force from the helical electromagnetic armature may provide the reaction force via electrical energy from the electrical energy source when the firing mechanism initiates movement of the projectile. To that point, a beginning of this reaction may precede, start simultaneously with, or follow the initiation of movement of the projectile.

(14) Another embodiment of the invention is an artillery broadly comprising a base, a firing mechanism, a guide rail, and a helical electromagnetic counter recoil mechanism having an internal configuration. The artillery may be configured to fire an external projectile such as a ring projectile. Other embodiments may encompass projectile propulsion devices more generally such as an electromagnetic launcher, a gas gun, a machine gun, or the like.

(15) The base supports or houses the firing mechanism and guide rail and may include a carriage, a gun mount, a firing platform, an elevator, and the like. The base may be maneuverable about a vertical axis (azimuth) and may be configured to pitch the guide rail about a horizontal axis (altitude).

(16) The firing mechanism activates an explosive charge of projectile and may be positioned aft of the guide rail. The firing mechanism may include a firing pin, a firing spring, a trigger, electrical or mechanical actuators, or the like.

(17) The guide rail extends forward from the firing mechanism and includes a central chamber configured to house the helical electromagnetic counter recoil mechanism. The guide rail is configured to be encircled by the projectile and to be aimed at a target. The guide rail may include rifling, muzzle brakes, propellant gas ports, and the like.

(18) The helical electromagnetic counter recoil mechanism is at least partially positioned in the central chamber and broadly comprises an electrical energy source, a positive rail, a negative rail, a number of brushes, a stator, and a helical electromagnetic armature. The helical electromagnetic counter recoil mechanism is at least partially encircled by the guide rail.

(19) The electrical energy source provides electrical energy to at least one of the stator and helical electromagnetic armature when the projectile is fired. The electrical energy source may be a battery, a capacitor, or other electrical energy sources.

(20) The positive rail electrically connects the stator to the electrical energy source via one of the brushes. Similarly, the negative rail electrically connects the helical electromagnetic armature to the electrical energy source via another one of the brushes.

(21) The stator includes helical coils and is stationary relative to the helical electromagnetic armature. The stator may be longitudinally aligned with the guide rail and encircles the helical electromagnetic armature. The stator may also be encircled by at least a portion of the guide rail for more compact construction. The stator is electrically connected to the positive rail via one of the brushes and to the helical electromagnetic armature via another one of the brushes.

(22) The helical electromagnetic armature includes helical coils and is configured to longitudinally translate relative to the stator. The helical electromagnetic armature may be encircled by at least a portion of the stator and at least a portion of the guide rail. The helical electromagnetic armature is

electrically connected to the stator via one of the brushes and to the negative rail via another one of the brushes.

(23) Another embodiment of the invention is a helical counter recoil mechanism broadly comprising a housing, an electrical energy source, a stator, and a helical electromagnetic armature. The helical counter recoil mechanism is configured to be integrated into or implemented with an actuation device or system including an actuatable component or any system or device that induces recoil (e.g., vibration, oscillation, or jerk). For example, the helical electromagnetic counter recoil mechanism may be used with a motor (e.g., an electric motor), an internal combustion engine, a turbine, a pump, or any other similar device or system that induces recoil. It should also be noted that “direction” as it relates to actuation, recoil, and counter recoil may be linear direction, rotational direction, or general direction (such as the case with poorly-defined machine vibrations). For exemplary purposes, the helical electromagnetic counter recoil mechanism will be described in conjunction with a pneumatic jackhammer including a compressor, a body, a compression chamber, a valve, an actuatable component, and a bit. The helical electromagnetic counter recoil mechanism may be an integral part of the jackhammer or may be an add-on component. To that end, the housing may be part of or secured to a body of the jackhammer and at least partially encloses the stator and the helical electromagnetic armature.

(24) The electrical energy source provides electrical energy to at least one of the stator and the helical electromagnetic armature when the actuatable component is actuated. The electrical energy source may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources.

(25) The stator may include helical coils and may be stationary relative to the helical electromagnetic armature. The stator may also be longitudinally and concentrically aligned with the helical electromagnetic armature.

(26) The helical electromagnetic armature may include helical coils and is configured to longitudinally translate relative to the stator. The helical electromagnetic armature may encircle or be encircled by at least a portion of the stator. The helical electromagnetic armature may also be electrically connected to the stator. The helical electromagnetic armature may also be configured to move in general opposition to movement of the actuatable component.

(27) In use, the helical electromagnetic counter recoil mechanism minimizes recoil of the jackhammer, in this case due to movement of the actuatable component in a first direction. More specifically, the jackhammer may be used as a conventional jackhammer, with the actuatable component being actuated toward the bit via compressed air from the compressor.

(28) The electrical energy source may then provide electrical energy to at least one of the stator and the helical electromagnetic armature to create one or more electromagnetic fields. This urges the helical electromagnetic armature in a second direction opposite the first direction, thereby minimizing recoil of the jackhammer from movement of the actuatable component in the first direction.

(29) The helical electromagnetic armature may then be reset to or close to its initial position (i.e., moved in the first direction) via an additional electromagnetic field or fields, thereby minimizing recoil of the jackhammer from movement of the actuatable component in the second direction. Thus, from the user's perspective, the jackhammer may vibrate or oscillate less violently or not at all, resulting in a significantly less fatiguing experience. This may also improve efficiency and accuracy since the user does not have to fight violent vibrations or oscillations.

(30) Another embodiment of the invention is a helical electromagnetic counter recoil mechanism configured to be used with a firearm. The helical counter recoil mechanism broadly comprises a housing, an electrical energy source, a stator, and a helical electromagnetic armature.

(31) The helical electromagnetic counter recoil mechanism is configured to be integrated into or removeably attached to the firearm. That is, the helical electromagnetic counter recoil mechanism may be modular, portable, or adaptable. To that end, the housing may be configured to be attached

to the barrel (or to the firearm near the barrel).

(32) The electrical energy source provides electrical energy to at least one of the stator and the helical electromagnetic armature when a projectile is propelled in the first direction. The electrical energy source may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources.

(33) The stator may include helical coils and may be stationary relative to the helical electromagnetic armature. The stator may also be longitudinally and concentrically aligned with the helical electromagnetic armature.

(34) The helical electromagnetic armature may include helical coils and is configured to longitudinally translate relative to the stator. The helical electromagnetic armature may encircle or be encircled by at least a portion of the stator. The helical electromagnetic armature may also be electrically connected to the stator. The helical electromagnetic armature may be configured to be reset, and in one embodiment automatically reset, to an initial position after the projectile has been fired.

(35) In use, the helical electromagnetic counter recoil mechanism may be slid onto the barrel from the barrel's distal end. If the muzzle is present, the muzzle may need to be removed beforehand and then reattached to the barrel.

(36) The firearm may then be aimed at a target, and its trigger may then be activated to propel the projectile out of the barrel in the first direction. The electrical energy source may then provide electrical energy to at least one of the stator and the helical electromagnetic armature to create one or more electromagnetic fields. This urges the helical electromagnetic armature in a second direction opposite the first direction, thereby minimizing recoil of the firearm from movement of the projectile in the first direction.

(37) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the present invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) Embodiments of the present invention are described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. 1 is a partial cutaway elevation view of an artillery constructed in accordance with an embodiment of the invention;

(3) FIG. 2 is an enlarged cutaway elevation view of a helical electromagnetic counter recoil mechanism of the artillery of FIG. 1;

(4) FIG. 3 is a partial cutaway elevation view of an artillery constructed in accordance with another embodiment of the invention;

(5) FIG. 4 an enlarged cutaway elevation view of a helical electromagnetic counter recoil mechanism of the artillery of FIG. 3;

(6) FIG. 5 is a partial cutaway elevation view of a helical electromagnetic counter recoil mechanism constructed in accordance with another embodiment of the invention;

(7) FIG. 6 is a partial cutaway perspective view of a helical electromagnetic counter recoil mechanism constructed in accordance with another embodiment of the invention.

(8) FIG. 7A is a schematic diagram of a helical electromagnetic counter recoil mechanism constructed in accordance with an embodiment of the invention;

(9) FIG. 7B is a schematic diagram of a helical electromagnetic counter recoil mechanism constructed in accordance with an embodiment of the invention;

(10) FIG. 7C is a schematic diagram of a helical electromagnetic counter recoil mechanism constructed in accordance with an embodiment of the invention; and

(11) FIG. 7D is a schematic diagram of a helical electromagnetic counter recoil mechanism constructed in accordance with an embodiment of the invention.

(12) The drawing figures do not limit the present invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(13) The following detailed description of the invention references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the present invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

(14) In this description, references to “one embodiment”, “an embodiment”, or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment”, “an embodiment”, or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the current technology can include a variety of combinations and/or integrations of the embodiments described herein.

(15) Turning to FIGS. 1 and 2, a projectile propulsion device **10** constructed in accordance with an embodiment of the invention is illustrated. The projectile propulsion device **10** broadly comprises a base **12**, a firing mechanism **14**, a barrel **16**, and a helical electromagnetic counter recoil mechanism **18**. The projectile propulsion device **10** may be configured to fire an internal projectile **100**. The projectile propulsion device **10** may be a tank gun, an airplane gun, a battleship gun, or the like.

(16) The base **12** supports or houses the firing mechanism **14** and barrel **16** and may include a carriage, a gun mount, a firing platform, an elevator; and the like. The base **12** may be maneuverable about a vertical axis (azimuth) and may be configured to pitch the barrel **16** about a horizontal axis (altitude).

(17) The firing mechanism **14** may be aft of the barrel **16** and activates an explosive charge **102** of the projectile **100**. The firing mechanism **14** may include a firing pin, a firing spring, a trigger, electrical or Mechanical actuators, or the like.

(18) The barrel **16** extends forward from the firing mechanism **14** and forms an open-ended bore **20**. The barrel **16** is configured to guide the projectile **100** out of the open end of the bore **20** towards a target. The barrel **16** may include rifling, muzzle brakes, propellant gas ports, and the like.

(19) The helical electromagnetic counter recoil mechanism **18** broadly comprises an electrical energy source **22**, a positive rail **24**, a negative rail **26**, a plurality of brushes **28A-D**, a stator **30**, and a helical electromagnetic armature **32**. The helical electromagnetic counter-recoil mechanism **18** at least partially encircles the barrel **16** so as to be external relative to the barrel. That is, the helical electromagnetic counter recoil mechanism **18** may be considered to have an “external” configuration. The helical electromagnetic counter recoil mechanism **18** may have a center of mass coaxially aligned with a center of mass of the barrel **16**.

(20) The electrical energy source **22** provides electrical energy to at least one of the stator **30** and

helical electromagnetic armature **32** via the positive rail **24**, negative rail **26**, and brushes **28A-D** when the firing mechanism **14** triggers the explosive charge **102** of the projectile **100**. The electrical energy source **22** may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources.

(21) The positive rail **24** is an electrical conduit electrically connecting the stator **30** to the electrical energy source **22** via brushes **28A,B**. To that end, the positive rail **24** may extend parallel to longitudinal axes of the stator **30** and helical electromagnetic armature **32** and the negative rail **26**.

(22) The negative rail **26** is an electrical conduit electrically connecting the helical electromagnetic armature **32** to the electrical energy source **22** via the brush **28D**. To that end, the negative rail **26** may extend parallel to the longitudinal axes of the stator **30** and helical electromagnetic armature **32** and the positive rail **24**.

(23) The brushes **28A-D** electrically connect the positive rail **24**, stator **30**, helical electromagnetic armature **32**, and negative rail **26** together while maintaining translatable freedom therebetween. Specifically, brush **28A** is configured to slideably engage the positive rail **24**. Brush **28B** is configured to slideably engage helical coils of the stator **30**. Brush **28C** is connected to helical coils of the helical electromagnetic armature **32** and is configured to slideably engage helical coils of the stator **30**. Brush **28D** is connected to the helical coils of the helical electromagnetic armature **32** opposite brush **28C** and is configured to slideably engage negative rail **26**.

(24) The stator **30** includes helical coils **34** and is stationary relative to the helical electromagnetic armature **32**. The stator **30** may be longitudinally aligned with the barrel **16**. The stator **30** may also encircle at least a portion of the barrel **16** for more compact construction. The stator **30** is electrically connected to the positive rail **24** via brushes **28A,B** and to the helical electromagnetic armature **32** via brush **28C**.

(25) The helical electromagnetic armature **32** includes helical coils **36** and is configured to longitudinally translate relative to the stator **30**. The helical electromagnetic armature **32** may encircle at least a portion of the stator **30** and may encircle at least a portion of the barrel **16**. The helical electromagnetic armature **32** is electrically connected to the stator **30** via brush **28C** and to the negative rail **26** via brush **28D**. The helical electromagnetic armature **32** may have a mass larger than a mass of the projectile **100** to minimize travel distance of the helical electromagnetic armature **32**. The helical electromagnetic armature **32** may be configured to be reset to an initial position after the projectile propulsion device **10** has fired.

(26) The above-described helical electromagnetic counter recoil mechanism **18** includes brush connections and a stator coil and armature coil configuration for channeling electricity through the stator **30** and the helical electromagnetic armature **32**. In this case, the stator **30** and helical electromagnetic armature **32** may impart an electromagnetic force on each other. Other helical electromagnetic configurations may also be used. For example, the helical electromagnetic counter recoil mechanism **18** may be configured so that electricity is channeled only through the stator or only through the armature. Other configurations such as separate electrical circuits for the stator **30** and helical electromagnetic armature **32** may be used.

(27) Use of the projectile propulsion device **10** will now be described in more detail. The projectile **100** may be loaded in the barrel **16** near an aft end of the bore **20** (e.g., in a loading chamber). The barrel **16** may then be aimed at a target or in the initial desired trajectory of the projectile **100**.

(28) The projectile **100** may then be fired at or toward the target. Specifically, the firing mechanism **14** may engage the explosive charge **102**. The explosive charge **102** may then detonate, in turn propelling the projectile **100** in a first direction (i.e., out the open end of the barrel **16**). Due to rapid forward acceleration of the projectile **100**, the projectile propulsion device **10** (or more specifically, the base **12** and barrel **16**) receives a reaction force realized in the form of recoil, which is minimized or eliminated by the helical electromagnetic counter recoil mechanism **18** as described below.

(29) Specifically, the electrical energy source **22** may provide electrical energy, in the form of

electrical current to the stator **30** and helical electromagnetic armature **32** via a circuit path routed through the positive rail **24**, brushes **28A-D**, and negative rail **26**. Current passing through the coils **34** of the stator **30** and/or the coils **36** of the helical electromagnetic armature **32** create an electromagnetic force that moves the helical electromagnetic armature **32** in a second direction opposite the first direction at substantially the same time the projectile **100** is propelled in the first direction. The projectile propulsion device **10** receives a reaction force from the helical electromagnetic armature **32**. In this way, the reaction force from the helical electromagnetic armature **32** counters the reaction force or recoil imparted on the projectile propulsion device **10** by the projectile **100**.

(30) The helical electromagnetic counter recoil mechanism **18** may then be reset. This may be done manually or via an application of reverse current or a reset mechanism.

(31) The above-described projectile propulsion device **10** provides several advantages. For example, the helical electromagnetic counter recoil mechanism **18** minimizes or eliminates recoil from firing projectiles. The helical electromagnetic armature **32** may be coaxially aligned with the barrel **16** and bore **20** or may have a center of mass aligned with a center of mass of the projectile **100** so that a moment or rotational force is not imparted on the projectile propulsion device **10**. Aspects of the helical electromagnetic counter recoil mechanism **18** can be easily adjusted to optimize performance. For example, the amount of current or a current profile can be changed via electronic circuitry, computer control, or the like to precisely control movement of the helical electromagnetic armature **32**. In this way, the exact amount of force over time exerted by firing the projectile can be countered to eliminate any impulse peak, delayed reaction, reverberation, secondary forces, natural frequencies, variations in charge power (e.g., imperfect munitions), different projectile gauges and powers, and the like. To that end, the helical electromagnetic armature **32** can even reverse direction during recoil mitigation. The helical electromagnetic counter recoil mechanism **18** may at least partially encircle the barrel **16**, which may reduce size of the projectile propulsion device **10**.

(32) Turning to FIGS. **3** and **4**, a projectile propulsion device **200** constructed in accordance with another embodiment of the invention is illustrated. The projectile propulsion device **200** broadly comprises a base **202**, a firing mechanism **204**, a guide rail **206**, and a helical electromagnetic counter recoil mechanism **208**. The projectile propulsion device **200** may be configured to fire an external projectile. To that end, a ring projectile **300** is shown, but other projectiles with female orientation may be used.

(33) The base **202** supports or houses the firing mechanism **204** and guide rail **206** and may include a carriage, a gun mount, a firing platform, an elevator, and the like. The base **202** may be maneuverable about a vertical axis (azimuth) and may be configured to pitch the guide rail **206** about a horizontal axis (altitude).

(34) The firing mechanism **204** activates an explosive charge **302** of projectile **300** and may be positioned aft of the guide rail **206**. The firing mechanism **204** may include a firing pin, a firing spring, a trigger, electrical or mechanical actuators, or the like.

(35) The guide rail **206** extends forward from the firing mechanism **204** and includes a central chamber **210** configured to house the helical electromagnetic counter recoil mechanism **208**. Alternatively, the central chamber **210** may be aft of the guide rail **206** in the base **202**. The guide rail **206** is configured to be encircled by the projectile **300** and to be aimed at a target (or directed to axially coincide with an initial trajectory of the projectile **300**) to guide the projectile **300** towards the target. The guide rail **206** may include rifling, muzzle brakes, propellant gas ports, and the like.

(36) The helical electromagnetic counter recoil mechanism **208** is at least partially positioned in the central chamber **210** and broadly comprises an electrical energy source **212**, a positive rail **214**, a negative rail **216**, a plurality of brushes **218A-D**, a stator **220**, and a helical electromagnetic armature **222**. The helical electromagnetic counter recoil mechanism **208** is at least partially encircled by the guide rail **206**. That is, the helical electromagnetic counter recoil mechanism **18**

may be considered to have an “internal” configuration. The helical electromagnetic counter recoil mechanism **18** may have a center of mass coaxially aligned with a center of mass of the guide rail **206**.

(37) The electrical energy source **212** provides electrical energy to at least one of the stator **220** and helical electromagnetic armature **222** via the positive rail **214**, negative rail **216**, and brushes **218A-D** when the firing mechanism **204** triggers the explosive charge **302** of the projectile **300**. The electrical energy source **212** may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources.

(38) The positive rail **214** is an electrical conduit electrically connecting the stator **220** to the electrical energy source **212** via brushes **218A,B**. To that end, the positive rail **214** may extend parallel to longitudinal axes of the stator **220** and helical electromagnetic armature **222** and the negative rail **216**.

(39) The negative rail **216** is an electrical conduit electrically connecting the helical electromagnetic armature **222** to the electrical energy source **212** via the brush **218D**. To that end, the negative rail **216** may extend parallel to the longitudinal axes of the stator **220** and helical electromagnetic armature **222** and the positive rail **214**.

(40) The brushes **218A-D** electrically connect the positive rail **214**, stator **220**, helical electromagnetic armature **222**, and negative rail **216** together while maintaining translatable freedom therebetween. Specifically, brush **218A** is configured to slideably engage the positive rail **214**. Brush **218B** is configured to slideably engage helical coils of the stator **220**. Brush **218C** is connected to helical coils of the helical electromagnetic armature **222** and is configured to slideably engage helical coils of the stator **220**. Brush **218D** is connected to the helical coils of the helical electromagnetic armature **222** opposite brush **218C** and is configured to slideably engage negative rail **216**.

(41) The stator **220** includes helical coils **224** and is stationary relative to the helical electromagnetic armature **222**. The stator **220** may be longitudinally aligned with the guide rail **206** and encircles the helical electromagnetic armature **222**. The stator **220** may also be encircled by at least a portion of the guide rail **206** for more compact construction. The stator **220** is electrically connected to the positive rail **214** via brushes **218A,B** and to the helical electromagnetic armature **222** via brush **218C**.

(42) The helical electromagnetic armature **222** includes helical coils **226** and is configured to longitudinally translate relative to the stator **220**. The helical electromagnetic armature **222** may be encircled by at least a portion of the stator **220** and at least a portion of the guide rail **206**. The helical electromagnetic armature **222** is electrically connected to the stator **220** via brush **218C** and to the negative rail **216** via brush **218D**. The helical electromagnetic armature **222** may be configured to be reset to an initial position after the projectile propulsion device **200** has fired.

(43) The above-described helical electromagnetic counter recoil mechanism **208** includes brush connections and a stator coil and armature coil configuration for channeling electricity through the stator **220** and the helical electromagnetic armature **222**. In this case, the stator **220** and helical electromagnetic armature **222** may impart an electromagnetic force on each other. Other helical electromagnetic configurations may also be used. For example, the helical electromagnetic counter recoil mechanism **208** may be configured so that electricity is channeled only through the stator or only through the armature. Other configurations such as separate electrical circuits for the stator **220** and helical electromagnetic armature **222** may be used.

(44) Use of the projectile propulsion device **200** will now be described in more detail. The projectile **300** may be loaded on the guide rail **206**. The guide rail **206** may then be aimed at a target or in the initial desired trajectory of the projectile **300**.

(45) The projectile **300** may then be fired at or toward the target. Specifically, the firing mechanism **204** may engage the explosive charge **302**. The explosive charge **302** may then detonate, thus propelling the projectile **300** in a first direction off a distal end of the guide rail **206**. Due to rapid

forward acceleration of the projectile **300**, the projectile propulsion device **200** (or more specifically, the base **202** and guide rail **206**) receives a reaction force realized in the form of recoil, which is minimized or eliminated by the helical electromagnetic counter recoil mechanism **208** as described below.

(46) Specifically, the electrical energy source **212** may provide electrical energy in the form of electrical current to the stator **220** and helical electromagnetic armature **222** via a circuit path routed through the positive rail **214**, brushes **218A-D**, and negative rail **216**. Current passing through the coils **224** of the stator **220** and the coils **226** of the helical electromagnetic armature **222** create an electromagnetic force that moves the helical electromagnetic armature **222** in a second direction opposite the first direction at substantially the same time the projectile **300** is propelled in the first direction. The projectile propulsion device **200** receives a reaction force from the helical electromagnetic armature **222**. In this way, the reaction force from the helical electromagnetic armature **222** counters the reaction force or recoil imparted on the projectile propulsion device **200** by the projectile **300**.

(47) The helical electromagnetic counter recoil mechanism **208** may then be reset. This may be done manually or via an application of reverse current or a reset mechanism.

(48) Turning to FIG. 5, a helical electromagnetic counter recoil mechanism **400** constructed in accordance with another embodiment of the invention is illustrated. The helical counter recoil mechanism **400** broadly comprises a housing **402**, an electrical energy source **404**, a stator **406**, and a helical electromagnetic armature **408**.

(49) The helical electromagnetic counter recoil mechanism **400** is configured to be integrated into or implemented with an actuation device or system including an actuatable component. For example, the helical electromagnetic counter recoil mechanism **400** may be used with a motor (e.g., an electric motor), an internal combustion engine, a turbine, a pump, or any other similar device or system that induces recoil (e.g., vibration, oscillation, or jerk). For exemplary purposes, the helical electromagnetic counter recoil mechanism **400** will be described in conjunction with a pneumatic jackhammer **500** including a compressor **502**, a body **504**, a compression chamber **506**, a valve **508**, an actuatable component **510**, and a bit **512**.

(50) The compressor **502** provides compressed air to the compression chamber **506**. The compressor **502** may be a tank or pump separate from the jackhammer **500** and pneumatically connected to the jackhammer **500** via pneumatic lines.

(51) The body **504** encloses the valve **508**, and the actuatable component **510**. The body may define the compression chamber **506** and a number of air channels leading to the compression chamber **506**. For example, the compressor **502** may be pneumatically connected to the compression chamber **506** via some of the air channels.

(52) The compression chamber **506** houses the actuatable component **510** and receives compressed air from the compressor **502** via the valve **508**. The compression chamber **506** may be pneumatically isolated via a number of seals.

(53) The valve **508** dictates airflow to the compression chamber **506** from the compressor **502** and may be configured to open and close cyclically or in reaction to movement of the actuatable component **510**. Opening of the valve **508** allows compressed air into the compression chamber **506**, which in turn actuates the actuatable component **510** in a first direction. The valve **508** may be closed upon the actuatable component **510** reaching a predetermined stroke point, which may urge the actuatable component **510** in a second direction opposite the first direction back to an initial stroke point.

(54) The actuatable component **510** is positioned in the compression chamber **506** and may be a piston configured to move within the compression chamber **506**. The actuatable component **510** may be configured to move in the first direction along an axis upon actuation toward the bit **512**. The actuatable component **510** may also be configured to strike the bit **512** upon reaching the predetermined stroke point.

(55) The bit **512** may be an elongated wedge or pin configured to strike a substrate **514** upon actuation of the actuatable component **510**. The bit **512** may thereby be configured to deconstruct or separate the substrate **514**.

(56) With respect again to the helical electromagnetic counter recoil mechanism **400**, the helical electromagnetic counter recoil mechanism **400** may be an integral part of the jackhammer **500** or may be an add-on component. To that end, the housing **402** may be part of or secured to the body **504** of the jackhammer **500** and at least partially encloses the stator **406** and the helical electromagnetic armature **408**.

(57) The electrical energy source **404** provides electrical energy to at least one of the stator **406** and the helical electromagnetic armature **408** when the actuatable component **510** is actuated. The electrical energy source **404** may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources. The electrical energy source **404** may be configured to be recharged via movement of the actuatable component **510** (or other mechanical energy).

(58) The stator **406** may include helical coils and may be stationary relative to the helical electromagnetic armature **408**. The stator **406** may also be longitudinally and concentrically aligned with the helical electromagnetic armature **408**:

(59) The helical electromagnetic armature **408** may include helical coils and is configured to longitudinally translate relative to, the stator **406**. The helical electromagnetic armature **408** may encircle or be encircled by at least a portion of the stator **406**. The helical electromagnetic armature **408** may also be electrically connected to the stator **406**. The helical electromagnetic armature **408** may be configured to be reset to an initial position after the actuatable component **510** has been actuated. The helical electromagnetic armature **408** may also be configured to move in general opposition to movement of the actuatable component **510**.

(60) The helical electromagnetic armature **408** may have a center of mass configured to be aligned with a center of mass of the actuatable component **510** in the first direction and the second direction so the actuatable component **510** and helical electromagnetic armature **408** do not impart rotation on the jackhammer **500**. The helical electromagnetic counter recoil mechanism **400** may be configured for actuatable components of a predetermined mass, with the helical electromagnetic armature **408** having a mass larger than the predetermined mass to minimize a travel velocity and a travel distance of the helical electromagnetic armature **408**.

(61) The above-described helical electromagnetic counter recoil mechanism **400** may include brush connections and a stator coil and armature coil configuration for channeling electricity through the stator **406** and the helical electromagnetic armature **408**. In this case, the stator **406** and helical electromagnetic armature **408** may impart an electromagnetic force on each other. Other helical electromagnetic configurations may also be used. For example, the helical electromagnetic counter recoil mechanism **400** may be configured so that electricity is channeled only through the stator **406** or only through the helical electromagnetic armature **408**. Other configurations such as separate electrical circuits for the stator **406** and helical electromagnetic armature **408** may be used.

(62) In use, the helical electromagnetic counter recoil mechanism **400** minimizes recoil of the jackhammer **500**, in this case due to movement of the actuatable component **510** in a first direction. More specifically, the jackhammer **500** may be used as a conventional jackhammer, with the actuatable component **510** being actuated toward the bit **512** via compressed air from the compressor **502**.

(63) The electrical energy source **404** may then provide electrical energy to at least one of the stator **406** and the helical electromagnetic armature **408** to create one or more electromagnetic fields. This urges the helical electromagnetic armature **408** in a second direction opposite the first direction, thereby minimizing recoil of the jackhammer **500** from movement of the actuatable component **510** in the first direction.

(64) The helical electromagnetic armature **408** may then be reset to or close to its initial position (i.e., moved in the first direction) via an additional electromagnetic field or fields, thereby

minimizing recoil of the jackhammer **500** from movement of the actuatable component **510** in the second direction. Thus, from the user's perspective, the jackhammer **500** may vibrate or oscillate less violently or not at all, resulting in a significantly less fatiguing experience. This may also improve efficiency and accuracy since the user does not have to fight violent vibrations or oscillations.

(65) The above-described helical electromagnetic counter recoil mechanism **400** has been described in terms of a pneumatic jackhammer. However, it can be used in conjunction with a motor (e.g., an electric motor), an internal combustion engine, a turbine, a pump, or any other mechanical system that has intended or biproduct oscillations, vibrations, jerks, or movement.

(66) Turning to FIG. **6**, a helical electromagnetic counter recoil mechanism **600** constructed in accordance with another embodiment of the invention is illustrated. The helical counter recoil mechanism **600** broadly comprises a housing **602**, an electrical energy source **604**, a stator **606**, and a helical electromagnetic armature **608**.

(67) The helical electromagnetic counter recoil mechanism **600** is configured to be integrated into or removeably attached to a firearm configured to propel a projectile. To that end, the helical electromagnetic counter recoil mechanism **600** may be modular, portable, or adaptable. For exemplary purposes, the helical electromagnetic counter recoil mechanism **600** will be described in conjunction with firearm **700**.

(68) The firearm **700** may be a shoulder rifle, hunting rifle, assault rifle, sniper rifle, or any other suitable firearm. The firearm **700** may include common attachments such as a scope, magazine, bipod, stock extension, silencer, muzzle (see muzzle **706** described below), carry handle, and the like.

(69) The stock **702** may include a grip and trigger, a magazine port, an attachment rail, and the like. The stock **702** may also include a safety, aiming sight, and the like.

(70) The barrel **704** extends forward from the stock **702** and defines a central channel for propelling a projectile in a first direction. The barrel increases accuracy of the projectile's trajectory. To that end, the barrel **704** may include rifling.

(71) The muzzle **706** may be positioned on a distal end of the barrel **704** and may help reduce noise, recoil, and kick-up (raising of the barrel). The muzzle **706** may also be removably attached to the barrel **704**.

(72) With respect again to the helical electromagnetic counter recoil mechanism **700**, the helical electromagnetic counter recoil mechanism **700** may be an add-on, modular, or portable component. To that end, the housing **602** may be configured to be attached to the barrel **704** (or to the firearm **700** near the barrel **704**) and at least partially encloses the stator **606** and the helical electromagnetic armature **608**.

(73) The electrical energy source **604** provides electrical energy to at least one of the stator **606** and the helical electromagnetic armature **608** when the projectile is propelled in the first direction. The electrical energy source **604** may be a battery, a pulse-forming network (PFN), a capacitor, or other electrical energy sources.

(74) The stator **606** may include helical coils and may be stationary relative to the helical electromagnetic armature **608**. The stator **606** may also be longitudinally and concentrically aligned with the helical electromagnetic armature **608**.

(75) The helical electromagnetic armature **608** may include helical coils and is configured to longitudinally translate relative to the stator **606**. The helical electromagnetic armature **608** may encircle or be encircled by at least a portion of the stator **606**. The helical electromagnetic armature **608** may also be electrically connected to the stator **606**. The helical electromagnetic armature **608** may be configured to be reset, and in one embodiment automatically reset, to an initial position after the projectile has been fired.

(76) The above-described helical electromagnetic counter recoil mechanism **600** may include brush connections and a stator coil and armature coil configuration for channeling electricity through the

stator **606** and the helical electromagnetic armature **608**. In this case, the stator **606** and helical electromagnetic armature **608** may impart an electromagnetic force on each other. Other helical electromagnetic configurations may also be used. For example, the helical electromagnetic counter recoil mechanism **600** may be configured so that electricity is channeled only through the stator **606** or only through the helical electromagnetic armature **608**. Other configurations such as separate electrical circuits for the stator **606** and helical electromagnetic armature **608** may be used.

(77) In use, the helical electromagnetic counter recoil mechanism **600** may be removeably attached to the firearm **700** (the firearm **700** may be useable without the helical electromagnetic counter recoil mechanism **600**). For example, the helical electromagnetic counter recoil mechanism **600** may be attached to the barrel **704** or another suitable component of the firearm **700**. In one embodiment, the helical electromagnetic counter recoil mechanism **600** may be slid onto the barrel **704** from the barrel's distal end. If the muzzle **706** is present, the muzzle **706** may need to be removed beforehand and then reattached to the barrel **704**. Alternatively, the helical electromagnetic counter recoil mechanism **600** may be an integrated component of the firearm **700**.

(78) The firearm **700** may then be aimed at a target, and its trigger may then be activated to propel the projectile out of the barrel **704** in the first direction: The electrical energy source **604** may then provide electrical energy to at least one of the stator **606** and the helical electromagnetic armature **608** to create one or more electromagnetic fields. This urges the helical electromagnetic armature **608** in a second direction opposite the first direction, thereby minimizing recoil of the firearm **700** from movement of the projectile in the first direction.

(79) The helical electromagnetic armature **608** may then be reset to or close to its initial position (i.e., moved in the first direction) manually or automatically (e.g., via an additional electromagnetic field or fields). This may be particularly useful for an automatic or semi-automatic firearm. Thus, from the user's perspective, the firearm **700** may vibrate or oscillate less violently or not at all, resulting in a significantly less fatiguing experience. This may also improve efficiency and accuracy since the user does not have to fight violent vibrations or oscillations.

(80) Turning to FIGS. 7A-D, the above-described helical electromagnetic counter recoil mechanisms (represented hereafter as helical electromagnetic counter recoil mechanism **800**) may be used in conjunction with a motor **802** (FIG. 7A), an internal combustion engine **804** (FIG. 7B), a turbine **806** (FIG. 7C) or a pump **808** (FIG. 8D) to counter recoil produced by such devices.

(81) Although the invention has been described with reference to the embodiments illustrated in the attached drawing figures, it is noted that equivalents may be employed and substitutions made herein without departing from the scope of the invention as recited in the claims.

Claims

1. A helical electromagnetic counter recoil mechanism for an actuation system including an actuatable component, the helical electromagnetic counter recoil mechanism comprising: an electrical energy source configured to provide electrical energy when the actuatable component is actuated in a first direction; a stator electrically connected to the electrical energy source, the stator being configured to generate a first electromagnetic field via some of the electrical energy; and a helical electromagnetic armature configured to generate a second electromagnetic field via some of the electrical energy to thereby move in a second direction opposite the first direction to minimize recoil from movement of the actuatable component in the first direction, the electrical energy source being configured to be recharged via motion of a component of the actuation system.
2. The helical electromagnetic counter recoil mechanism of claim 1, the helical electromagnetic armature having a center of mass configured to be aligned with a center of mass of the actuatable component in the first direction and the second direction so the actuatable component and helical electromagnetic armature do not impart rotation on the actuation system.
3. The helical electromagnetic counter recoil mechanism of claim 1, the helical electromagnetic

armature being configured to be reset after the actuatable component has been actuated.

4. The helical electromagnetic counter recoil mechanism of claim 3, the helical electromagnetic armature being further configured to be reset via additional electrical energy from the electrical energy source.

5. The helical electromagnetic counter recoil mechanism of claim 1, the helical electromagnetic counter recoil mechanism being configured for actuatable components of a predetermined mass, the helical electromagnetic armature having a mass larger than the predetermined mass to minimize a travel velocity and a travel distance of the helical electromagnetic armature.

6. The helical electromagnetic counter recoil mechanism of claim 1, the electrical energy source being a pulse-forming network (PFN) configured to store the electrical energy and discharge the electrical energy to the stator.

7. The helical electromagnetic counter recoil mechanism of claim 1, the electrical energy source being configured to discharge some of the electrical energy to the helical electromagnetic armature, the helical electromagnetic armature being configured to impart an electromagnetic force on the stator.

8. The helical electromagnetic counter recoil mechanism of claim 1, the actuation system being selected from the group consisting of a motor, an internal combustion engine, a turbine, a pump, and a jackhammer.

9. The helical electromagnetic counter recoil mechanism of claim 1, further comprising a housing at least partially enclosing the stator and the electromagnetic armature.

10. A helical electromagnetic counter recoil mechanism for a recoil-inducing system, the helical electromagnetic counter recoil mechanism comprising: an electrical energy source configured to provide electrical energy when the recoil-inducing system induces recoil in a first direction; a stator electrically connected to the electrical energy source, the stator being configured to generate a first electromagnetic field via some of the electrical energy; and a helical electromagnetic armature configured to generate a second electromagnetic field via some of the electrical energy to thereby move in a second direction opposite the first direction to minimize the recoil, the electrical energy source being configured to be recharged via motion of a component of the recoil-inducing system.

11. A helical electromagnetic counter recoil mechanism for an actuation system including an actuatable component, the actuation system being selected from the group consisting of a motor, an internal combustion engine, a turbine, a pump, and a jackhammer, the helical electromagnetic counter recoil mechanism comprising: an electrical energy source configured to provide electrical energy when the actuatable component is actuated in a first direction; a stator electrically connected to the electrical energy source, the stator being configured to generate a first electromagnetic field via some of the electrical energy; and a helical electromagnetic armature configured to generate a second electromagnetic field via some of the electrical energy to thereby move in a second direction opposite the first direction to minimize recoil from movement of the actuatable component in the first direction.

12. The helical electromagnetic counter recoil mechanism of claim 11, the actuation system being a motor.

13. The helical electromagnetic counter recoil mechanism of claim 11, the actuation system being an internal combustion engine.

14. The helical electromagnetic counter recoil mechanism of claim 11, the actuation system being a turbine.

15. The helical electromagnetic counter recoil mechanism of claim 11, the actuation system being a pump.

16. The helical electromagnetic counter recoil mechanism of claim 11, the actuation system being a jackhammer.
